

Samuel Mathew

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EDUCATION

University of Waterloo

Bachelor of Computer Science (Hons.) | Specialization: Business Strategy

Sep. 2022 – Dec. 2026

Waterloo, Canada

TECHNICAL SKILLS

Languages: Python, Java, C++, C#, C, TypeScript, JavaScript, HTML, CSS, Kotlin, Bash.

Developer Tools: Git, Gitlab, Azure, IntelliJ IDEA, PyCharm, Eclipse, Jira, VS Code, Figma.

Libraries/Frameworks: React.js, GSAP, Node.js, Tailwind CSS, Bootstrap, Pandas, Selenium.

WORK EXPERIENCE

Sheridan College

Software Developer Intern

Jan. 2024 – Apr. 2024

Oakville, ON

- Engineered dynamic service pages and interactive UI components used by **500+** users via **Javascript** and **GSAP**.
- Developed **UML diagrams** in **Figma** to support the merge of **Zoho Desk** ticketing system ensuring readability.
- Eliminated manual bottlenecks by transitioning to an automated ticketing system, boosting workflow efficiency.
- Learned and mapped requirements for a new ticketing platform, evaluating functionality against project needs.
- Created **Articulate Rise 360** onboarding modules to streamline training and accelerate onboarding for new staff.
- Enhanced platform accessibility by collaborating with designers, leading to a **30%** reduction in reported **UI** issues.
- Contributed to the creation and curation of digital assets for the SPARK website enhancing visual consistency.

Kitchen Stuff Plus

Sales Associate

Nov. 2020 – May, 2026

Mississauga, ON

- Used an inventory system to track stock levels and submit restocking request across high-volume retail operations.
- Assisted **50–100** customers per shift by recommending products, leading to up to **\$50,000** in daily sales.
- Handled inventory management and POS transactions with **0%** error rates in a fast-paced retail environment.
- Trained new employees on internal systems and digital tools via clear instructions to ensure accurate system usage.

PROJECTS

Spotly 🌀 | *React.js, Typescript, Tailwind CSS, PostgreSQL, Node.js, Express.js, REST APIs, Figma*

- Secured all routes by building a **JWT Auth** flow in **TypeScript/Node.js** using a custom AuthStorage module.
- Enabled spot-saving by engineering a REST API collections system in **PostgreSQL/Node.js** using React hooks.
- Powered discovery by integrating SerpAPI into the **Express.js** backend, rendering results in React/Tailwind CSS.
- Reduced redundant API calls by building a PostgreSQL geo-cache in **Node.js/TypeScript** within a 75 km radius.

NutriGuard 🌀 | *Kotlin, Supabase, REST APIs, Figma*

- Constructed backend services for a food-tracking application using a **Supabase** infrastructure to support workflows.
- Designed database schemas and **Kotlin** services to ensure real-time inventory updates with strong data consistency.
- Built a unit and integration **test suite** that eliminated race conditions and prevented data-loss bugs prior to launch.
- Collaborated with **3** teammates using **GitLab** and bi-weekly sprints, delivering the tasks ahead of schedule.
- Integrated **Google ML Kit** for barcode scanning and **OpenFoodFacts API** for automated nutritional analysis.

Watopoly 🌀 | *C++, Makefile*

- Developed backend systems in **C++** for Monopoly by implementing player actions, and turn sequencing via **OOP**.
- Designed an in-memory architecture using **STL** containers to manage **100%** of game entities efficiently and safely.
- Integrated a **Observer Pattern** to update board state in real time, improving game clarity and debugging speed.
- Verified memory correctness using **Valgrind** to detect and eliminate leaks and invalid accesses across the codebase.

SPARK 🌀 | *JavaScript, HTML, TailwindCSS, GSAP.js*

- Architected a scalable, multi-level site using **HTML** and **JavaScript**, improving maintainability.
- Instituted consistent page layouts and routing-like navigation patterns reducing technical debt across the codebase.
- Built modular, reusable content sections using **Tailwind CSS** and **GSAP** animations, enabling rapid iteration.
- Implemented responsive design and accessibility-conscious markup to ensure consistent usability across devices.